

Problem Set 8

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1 Questions and Responses

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The closed-form estimates are extremely close to the true β values: $\hat{\beta}_{OLS}$ closely recovers the true parameter vector, with only very small deviations.

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The L-BFGS and Nelder-Mead estimates are nearly identical, but L-BFGS matches the closed-form OLS solution more precisely.

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The estimates of $\hat{\beta}$ are very similar to the true β : OLS, gradient descent, L-BFGS, and MLE are essentially identical, and Nelder-Mead is also extremely close. Across coefficients, the estimated values differ from the true values by a few thousandths.

Table 1: Comparison of OLS Estimates

	True β	Closed-Form	Gradient Descent	L-BFGS	Nelder-Mead	MLE
X_1	1.500	1.501	1.501	1.501	1.501	1.501
X_2	-1.000	-1.001	-1.001	-1.001	-1.001	-1.001
X_3	-0.250	-0.252	-0.252	-0.252	-0.252	-0.252
X_4	0.750	0.749	0.749	0.749	0.749	0.749
X_5	3.500	3.501	3.501	3.501	3.501	3.501
X_6	-2.000	-2.001	-2.001	-2.001	-2.001	-2.001
X_7	0.500	0.499	0.499	0.499	0.499	0.499
X_8	1.000	1.003	1.003	1.003	1.003	1.003
X_9	1.250	1.247	1.247	1.247	1.247	1.247
X_{10}	2.000	2.001	2.001	2.001	2.001	2.001
σ	0.500					0.500

Table 2: OLS Estimates

	(1)
X_1	1.501 (0.002)
X_2	-1.001 (0.002)
X_3	-0.252 (0.002)
X_4	0.749 (0.002)
X_5	3.501 (0.002)
X_6	-2.001 (0.002)
X_7	0.499 (0.002)
X_8	1.003 (0.002)
X_9	1.247 (0.002)
X_{10}	2.001 (0.002)
Observations	100,000
R^2	0.991
Adj. R^2	0.991

Notes: This table reports coefficients from the easy OLS regression of Y on X_1 through X_{10} . Standard errors are reported in parentheses. All coefficients are statistically significant at the 1 percent level.